

Hiring Management & Technical Leadership Team

Automotive Software Development & Infotainment Systems

Subject: Application for Senior Technical Lead / Software Architect – Automotive HMI & Embedded Systems

Dear Hiring Manager,

I am writing to express my strong interest in senior technical leadership opportunities within Automotive HMI, Embedded Software, and Infotainment Systems Architecture. With over 13 years of experience specializing in C++, Qt/QML, OpenGL ES, and Embedded Linux—highlighted by nearly a decade architecting next-generation production software for Mercedes-Benz MBUX and MB.OS platforms—I bring a proven track record of bridging high-fidelity UX design and high-performance embedded systems.

Throughout my career, I have specialized in tackling complex graphics engineering and interactive UI challenges on resource-constrained hardware. Currently, as an external consultant at Xpheno (client MBRDI), I lead the ownership and real-time performance optimization of an in-house 3D automotive visualization engine built on Qt and custom GLSL shaders. My role encompasses driving GPU-accelerated rendering effects, refining scene graph architecture, and guiding migration strategies toward Unity-based rendering architectures for future Software Defined Vehicle (SDV) platforms.

Key Architectural & Technical Accomplishments Include:

- **MB.OS & MBUX Production Frameworks:** Architected and developed modular, reusable UI control frameworks using modern C++ and Qt6/QML, establishing design patterns deployed across high-volume production vehicle models.
- **Conversational AI & VUI Systems:** Spearheaded an in-house Sound Dialogue System (SDS) prototype featuring dynamic runtime switching across cloud NLU providers (Dialogflow, Rasa, Wit.ai, Houndify) to evaluate internal voice platforms.
- **Graphics & GPU Optimization:** Built shader-driven dynamic effects, optimized rendering pipelines, and solved memory/execution bottlenecks across Linux-based automotive ECUs.
- **Engineering Leadership:** Led technical mentoring, code reviews, and cross-site architectural alignment between software teams in Germany and India.

What sets my approach apart is my comprehensive lifecycle experience—spanning early-stage research prototypes and NLU integration to full production rollout, technical mentorship, and multi-team collaboration. Recognized with awards such as the Mercedes-Benz Innovation Boot Camp Winner (2019) and Silver Star (2020), I thrive in environments that demand technical rigor, creative problem-solving, and continuous innovation in the automotive domain.

I would welcome the opportunity to discuss how my technical leadership, 3D/HMI architecture experience, and deep domain knowledge in automotive software can add immediate value to your organization's engineering roadmap. Thank you for your time and consideration.

Sincerely,

Thoufiq Yahya